

An introduction version control

https://stirlingcodingclub.github.io/version_control.pdf

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13 January 2023

Using version control

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- ▶ Focus on practical skills for research
 - ▶ Learn and reinforce knowledge on how to use **key skills** effectively
 - ▶ Focus on [GitHub](#) and [GitHub Desktop](#) software

Using version control

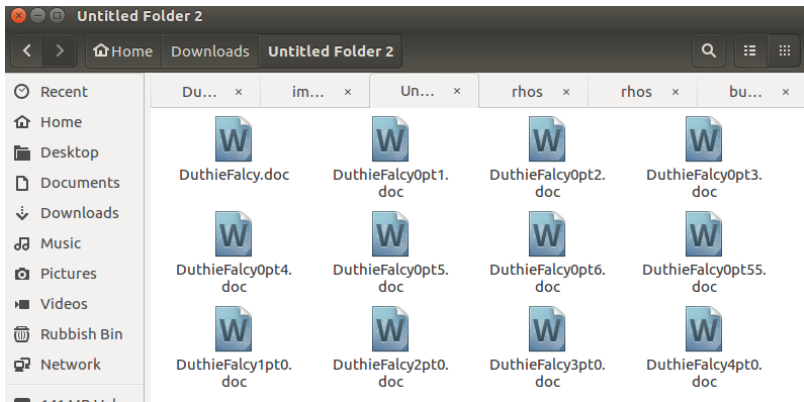
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- ▶ Focus on practical skills for research
 - ▶ Learn and reinforce knowledge on how to use **key skills** effectively
 - ▶ Focus on [GitHub](#) and [GitHub Desktop](#) software
- ▶ Hands-on practice setting up and using version control in your own work with [accompanying notes for guidance](#)

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Rough outline of version control

1. What is version control, and why use it?
2. Getting set up – good file management
3. Setting up [GitHub](#)
4. The [GitHub Desktop](#) tutorial
5. Independent work using version control

What is version control, and why use it?



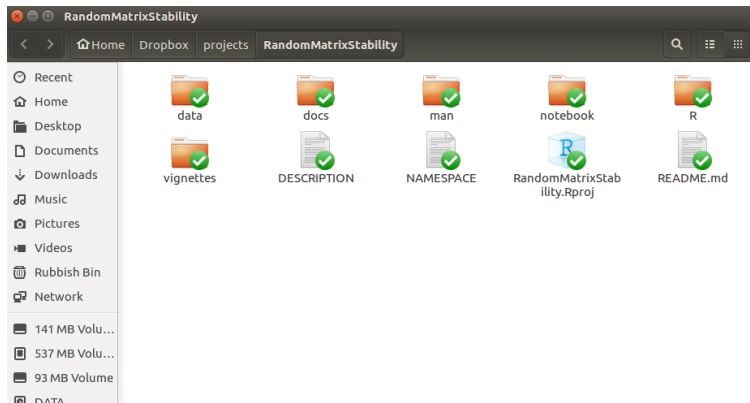
What version control software does

- ▶ Software that records changes you make to files over time
 - ▶ Manage different *versions* of files (no need to 'Save As...')
 - ▶ Recover old files, keep track of file changes
 - ▶ Collaborate with others on shared files

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- ▶ **Put more intuitively**, version control takes a snapshot in time (called a '**commit**') of all the files in one of your folders (called '**repositories**')
 - ▶ Visualise changes to your files over time
 - ▶ Look at the differences between file versions
 - ▶ Record who changed files, and what they changed

Inside of a project on version control



Folders (a.k.a, 'repositories') include all data files, R code, notes, manuscript drafts, etc.

Full annotated timeline of folder changes (GitHub)

 Commits on Mar 1, 2019

Some work on the SI
 bradduthie committed on 1 Mar 2019 ✓

 5d29331 

Major restructure and revision of the Discussion to compare to Gibbs
 bradduthie committed on 1 Mar 2019 ✓
...et al.

 a40ede5 

 Commits on Feb 27, 2019

Edit the manuscript up to Reviewer 2 specific comment 3 -- these are
 bradduthie committed on 27 Feb 2019 ✓
...next on the list, specifically the new Discussion paragraph

 d411fd5 

 Commits on Feb 22, 2019

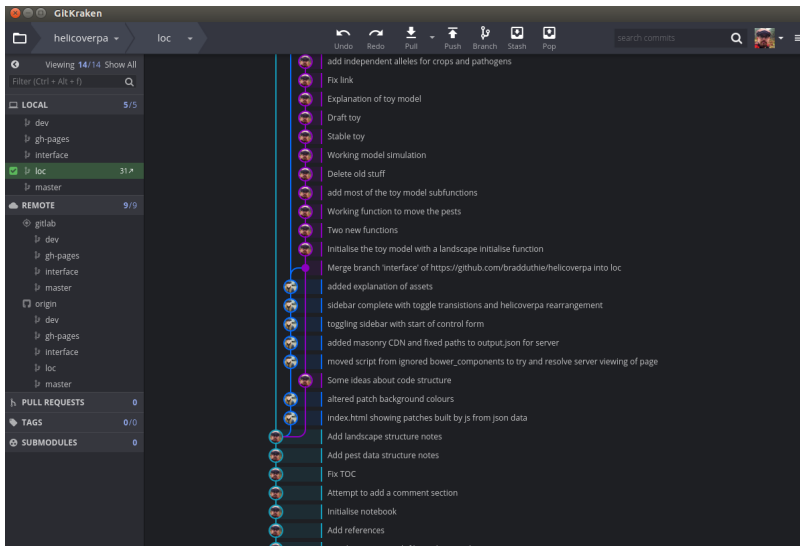
First pass, fix some notation
 bradduthie committed on 22 Feb 2019 ✓

 39f854b 

Add some to the abstract to emphasise finite systems
 bradduthie committed on 22 Feb 2019 ✓

 612aa4b 

Parallel versions ('branches') of a folder



Clear breakdown of what has changed

The screenshot shows a GitHub commit diff interface. At the top, it indicates "1 commit", "1 file changed", and "1 contributor". Below this, it shows the commit message "New line" by user "bradduthie" committed 4 minutes ago. The diff view shows a single file, "README.md", with 2 additions and 0 deletions. The diff is displayed in a dark-themed editor with line numbers. The original content (lines 3-5) is shown in a lighter blue background, and the new content (lines 6-7) is shown in a darker green background. The new content includes a line starting with a plus sign and the text "Adding a new line!".

1 commit 1 file changed 1 contributor

Commits on Nov 15, 2022

New line new
bradduthie committed 4 minutes ago

Showing 1 changed file with 2 additions and 0 deletions. Split Unified

2 README.md

@@ -3,3 +3,5 @@

3 This is your README. READMEs are where you can communicate what your project is and how to use it.

4

5 Write your name on line 6, save it, and then head back to GitHub Desktop.

6 +

7 + Adding a new line!

Version control makes science easier

- ▶ **Organises files** by avoiding 'save as' multiple versions
 - ▶ analysis_1.R
 - ▶ analysis_2.R
 - ▶ analysis_FINAL.R
 - ▶ analysis_FINAL_no_really_this_time.R

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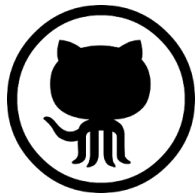
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- ▶ **Gives peace of mind** to experiment by removing any fear of breaking something that you know works

Version control can help open science



- ▶ Transparent record of data collection, analysis, and writing
- ▶ Record publicly available on [GitHub](#), [Bitbucket](#), or [GitLab](#)
- ▶ GitHub repository can be copied, reproduced, and discussed
- ▶ [git](#) and GitHub can track individual contributions to a project

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- ▶ Works across platforms
 - ▶ Windows
 - ▶ Linux
 - ▶ Mac
- ▶ Invented by [Linus Torvalds](#)